

EXPERIMENT NO. 3 — CLASSIFICATION ALGORITHM

Learning Objective: To implement a Classification algorithm using a Decision Tree (ID3/C4.5).

Tools: Python / Java / C / C++ / Data Mining Tool **Algorithm:** Decision Tree using ID3/C4.5.

1. Classification — Theory

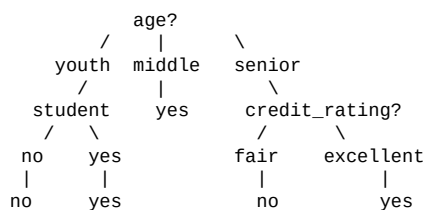
Classification is a supervised machine learning and data-mining technique used to assign an observation to one of a set of predefined classes. It learns a model from a labelled training dataset, where the input attributes and the correct class are already known. The learned model is then used to predict the class of new, unseen records.

For example, an online store may want to predict whether a customer will buy a computer. The input attributes can include **age, student status and credit rating**, while the target class is **buys_computer = yes/no**. The algorithm learns relationships between these attributes and the target class from historical examples.

Classification differs from clustering because classification is **supervised**: the training data contains class labels. In clustering, the algorithm discovers groups without predefined labels.

2. Decision Tree

A **Decision Tree** is a tree-shaped classification model. Each internal node represents a test on an attribute, each branch represents an outcome of that test, and each leaf node represents a predicted class. The topmost node is called the **root**.



A new record is classified by starting at the root and following the branch corresponding to the record's attribute value until a leaf is reached. The class stored at that leaf becomes the prediction.

3. Important Components

Component	Meaning
Root node	First and most important attribute selected for splitting the dataset.
Internal node	An attribute test used to divide the records.
Branch	One possible outcome/value of an attribute.
Leaf node	Final class label or prediction.
Training set	Labeled examples used to construct the tree.
Test/new tuple	Unseen record whose class must be predicted.

4. ID3 and C4.5

ID3 is a classic decision-tree algorithm that commonly selects the attribute with the highest **Information Gain**. Information Gain is based on entropy, which measures the impurity or uncertainty of a dataset.

C4.5 extends the basic ID3 idea and uses **Gain Ratio**, handles continuous attributes through thresholding, and can deal more effectively with missing values and tree pruning. Both algorithms are used to build decision trees by recursively selecting useful attributes.

The experiment sheet specifies ID3/C4.5; the implementation below demonstrates the core **ID3-style Information Gain** approach.

5. Entropy

Entropy measures the uncertainty or impurity of a collection of labeled records. If all records belong to the same class, entropy is 0 because there is no uncertainty. If classes are mixed, entropy is higher.

$$\text{Entropy}(S) = - \sum p_i \log_2(p_i)$$

where p_i is the proportion of records belonging to class i .

For a binary target with equal numbers of Yes and No:

$$\text{Entropy} = -(0.5 \log_2 0.5) - (0.5 \log_2 0.5)$$

$$= 1$$

For a pure set such as 5 Yes and 0 No:

$$\text{Entropy} = -(1 \log_2 1) = 0$$

CLASSIFICATION — DECISION TREE THEORY

6. Information Gain

Information Gain measures how much uncertainty is reduced when the dataset is split using an attribute. An attribute that creates purer subsets generally has higher Information Gain.

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum (|S_v| / |S|) \text{Entropy}(S_v)$$

S = complete dataset

A = selected attribute

S_v = subset for an attribute value v

The attribute with the largest Information Gain is selected as the next decision node in an ID3-style tree.

7. Recursive Tree Construction

Decision-tree induction is recursive. At the current node, the algorithm calculates the information gain of the available attributes, chooses the best attribute, divides the data according to its values, and repeats the process for every subset. Recursion stops when a subset is pure, when there are no useful attributes left, or when another stopping condition is reached.

```
Dataset
↓
Calculate Entropy
↓
Calculate Information Gain
↓
Select best attribute
↓
Split dataset
↓
Build child nodes recursively
↓
Pure subset / no attributes?
  ■■■ No → continue splitting
  ■■■ Yes → create leaf
```

8. Stopping Conditions

Typical stopping conditions are:

- All records in a node belong to the same class.
- No attributes remain for further splitting.
- A practical implementation may also stop using limits such as maximum depth or minimum samples.
- When no attribute provides useful separation, the majority class can be used as the leaf prediction.

9. Advantages

- Easy to understand and explain using a tree or rules.
- Can handle categorical attributes naturally.
- Little preprocessing is required compared with distance-based algorithms.
- The resulting tree can be converted into human-readable IF–THEN rules.
- Useful for exploratory analysis and classification.

10. Limitations

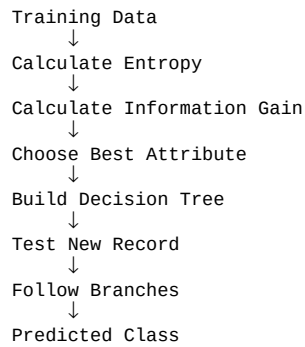
- A tree can overfit if it grows too deeply.
- Small changes in training data can sometimes produce a different tree.
- Greedy attribute selection does not always produce a globally optimal tree.
- Very large trees can become difficult to interpret.
- ID3's information gain can favor attributes with many distinct values.

11. Applications

- Medical diagnosis and disease classification.

- Credit approval and financial risk analysis.
- Customer purchase prediction.
- Fraud detection.
- Student performance classification.
- Email/spam classification and document categorization.

12. Classification Process



PART B — SAMPLE EXAMPLE: ALL ELECTRONICS

Consider the classic **AllElectronics** dataset. The objective is to predict whether a customer will **buy a computer** based on age, student status and credit rating.

13. Training Dataset

Age	Student	Credit Rating	Buys Computer
youth	no	fair	no
youth	no	excellent	no
middle_aged	no	fair	yes
senior	no	fair	yes
senior	no	fair	yes
senior	yes	excellent	no
middle_aged	yes	excellent	yes
youth	no	fair	no
youth	yes	fair	yes
senior	yes	fair	yes
youth	yes	excellent	yes
middle_aged	no	excellent	yes
middle_aged	yes	fair	yes
senior	no	excellent	no

There are **14 records**: 9 records have class **Yes** and 5 records have class **No**.

14. Entropy of the Complete Dataset

$$\begin{aligned}p(\text{Yes}) &= 9/14 \\ p(\text{No}) &= 5/14\end{aligned}$$

$$\begin{aligned}\text{Entropy}(S) &= -(9/14) \log_2(9/14) - (5/14) \log_2(5/14) \\ &\approx 0.940\end{aligned}$$

The dataset is therefore not pure because it contains both Yes and No classes.

15. Information Gain for Age

Split the dataset according to the Age attribute.

Age value	Records	Yes	No	Entropy
youth	5	2	3	≈ 0.971
middle_aged	4	4	0	0
senior	5	3	2	≈ 0.971

$$\begin{aligned}\text{Weighted entropy}(\text{Age}) &= (5/14)(0.971) + (4/14)(0) \\ &\quad + (5/14)(0.971) \\ &\approx 0.694\end{aligned}$$

$$\begin{aligned}\text{Gain}(S, \text{Age}) &= 0.940 - 0.694 \\ &\approx 0.246\end{aligned}$$

16. Choosing the Root

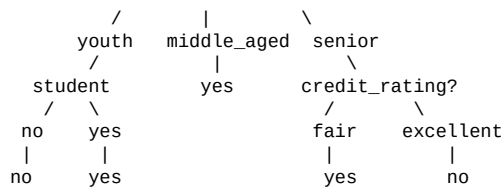
The algorithm calculates Information Gain for **Age, Student and Credit Rating**. For this dataset, Age gives the highest Information Gain, so **Age becomes the root node** in the classic ID3 tree.

Attribute	Approx. Information Gain
Age	0.246
Student	0.151
Credit Rating	0.048

17. Continue Splitting

The **middle_aged** subset contains only Yes records, so it becomes a leaf immediately. The **youth** and **senior** subsets still contain mixed classes, so the algorithm chooses another useful attribute inside each subset.

age?



This tree predicts the target class by following the path determined by the new customer's attribute values.

18. Sample Prediction 1

Customer:

Attribute	Value
Age	youth
Student	yes
Credit Rating	fair

age = youth
 ↓
 student = yes
 ↓
 Prediction = YES

Therefore, the model predicts that the customer **will buy a computer**.

19. Sample Prediction 2

Customer:

Attribute	Value
Age	senior
Student	no
Credit Rating	excellent

age = senior
 ↓
 credit_rating = excellent
 ↓
 Prediction = NO

Therefore, the model predicts that the customer **will not buy a computer**.

20. IF–THEN Rules

IF age = youth AND student = no
 THEN buys_computer = no

IF age = youth AND student = yes
 THEN buys_computer = yes

IF age = middle_aged
 THEN buys_computer = yes

IF age = senior AND credit_rating = fair
 THEN buys_computer = yes

IF age = senior AND credit_rating = excellent
 THEN buys_computer = no

21. Result and Conclusion

The Decision Tree classification algorithm was studied using the AllElectronics dataset. Entropy and Information Gain were used to select useful attributes, and a tree was constructed recursively. The resulting tree can classify new customers into **buys_computer = yes** or **no** by following the appropriate branches.

Decision trees are useful because the final model is easy to visualize and interpret. The same tree can also be converted into simple IF–THEN classification rules.